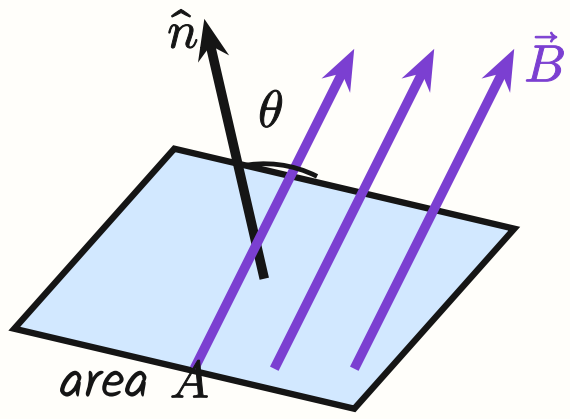
Magnetic Flux

Class 12 · UNIFORM $\vec{B}$ unless stated · $\otimes$ $\vec{B}$ into the page, $\odot$ out of the page · IDEAL inductors (no resistance) unless stated · t from the switching instant · $\mu_0 = 4\pi \times 10^{-7}\,\mathrm{T\,m\,A^{-1}}$

1 Flux through a flat area



$$\Phi = \vec{B}\cdot\vec{A} = BA\cos\theta$$

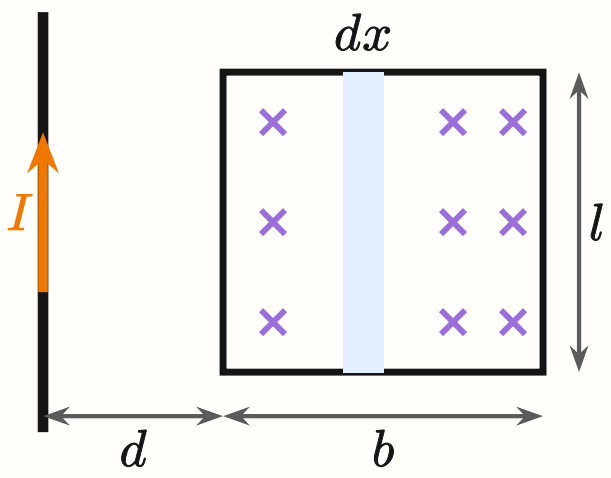
UNIFORM $\vec{B}$, FLAT area · θ = angle between $\vec{B}$ and the NORMAL $\hat{n}$ · unit weber: $1\,\mathrm{Wb} = 1\,\mathrm{T\,m^2}$ · scalar, sign set by the chosen $\hat{n}$

- Plane $\perp \vec{B}$ ($\theta = 0$): $\Phi = BA$ (maximum). Plane parallel to $\vec{B}$ ($\theta = 90°$): $\Phi = 0$.
- Coil of N tightly wound identical turns: flux linkage $N\Phi$. Closed surface: $\oint \vec{B}\cdot d\vec{A} = 0$ always.

Example

$B = 0.5\,\mathrm{T}$, $A = 0.02\,\mathrm{m^2}$, $\theta = 60°$:
$\Phi = 0.5 \times 0.02 \times 0.5 = 0.005\,\mathrm{Wb} = 5\,\mathrm{mWb}$.

2 Non-uniform field: integrate



$$\Phi = \int \vec{B}\cdot d\vec{A} = \frac{\mu_0 I l}{2\pi}\ln\frac{d+b}{d}$$

LONG straight wire, rectangle in the SAME plane · strip $dA = l\,dx$ at distance x, $B = \frac{\mu_0 I}{2\pi x}$ (into the page on the right of an upward current)

Syllabus Checklist

JEE Main + JEE Advanced + NEET syllabus + standard results

1 JEE Main + JEE Advanced syllabus

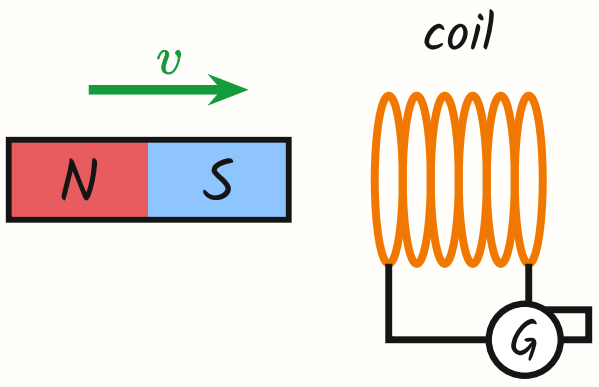
- ✓ Electromagnetic induction; magnetic flux — p1
- ✓ Faraday's law, induced EMF and current — p3
- ✓ Lenz's law (energy conservation) — p4
- ✓ Eddy currents and their applications — p13, p14
- ✓ Self-inductance (solenoid, toroid) — p15, p16
- ✓ Mutual inductance, coefficient of coupling — p17
- ✓ LR circuits with a d.c. source: growth and decay — p20–p22
- ✓ LC circuit: oscillations (JEE Advanced list) — p23, p24
- ✓ AC generator principle — p10

2 Also covered: hidden topics + standard results

- ✓ Motional EMF: rod on rails, force, power — p5, p6
- ✓ Arbitrary-shaped conductor; rod in a non-uniform field — p7
- ✓ Rotating rod and rotating disc — p8
- ✓ Loop crossing a field region (graphs) — p9
- ✓ Induced electric field, $\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi}{dt}$ — p11
- ✓ Cylindrical region: field inside and outside — p11, p12
- ✓ Charge flow $q = N\Delta\Phi/R$; back-EMF graphs — p3, p16
- ✓ Series and parallel inductors — p18
- ✓ Energy in an inductor, energy density $\frac{B^2}{2\mu_0}$ — p19
- ✓ Special cases, results, traps, solved questions — p25–p33

Faraday's Laws

1 Faraday's experiments



- Magnet moved towards or away from the coil: deflection ONLY while it moves.
- Current in a nearby coil switched on or off: momentary current; steady current: none.

2 The law

$$\varepsilon = -N\frac{d\Phi}{dt} \qquad \varepsilon_{avg} = -N\frac{\Delta\Phi}{\Delta t}$$

N turns, each linking Φ · minus sign = Lenz's law · Φ changes through B, the AREA or θ · total resistance R: $I = \varepsilon/R$

3 Charge that flows

$$q = \int I\,dt = \frac{N|\Delta\Phi|}{R}$$

closed circuit, TOTAL resistance R, no inductance effects · independent of HOW FAST the flux changes

Example

100 turns, area $0.05\,\mathrm{m}^2$, $\vec{B} \perp$ plane falls from $0.4\,\mathrm{T}$ to zero in $0.5\,\mathrm{s}$ (uniformly): $\varepsilon = 100 \times 0.05 \times 0.4/0.5 = 4\,\mathrm{V}$; with $R = 20\,\Omega$: $q = 100 \times 0.05 \times 0.4/20 = 0.1\,\mathrm{C}$.

Lenz's Law

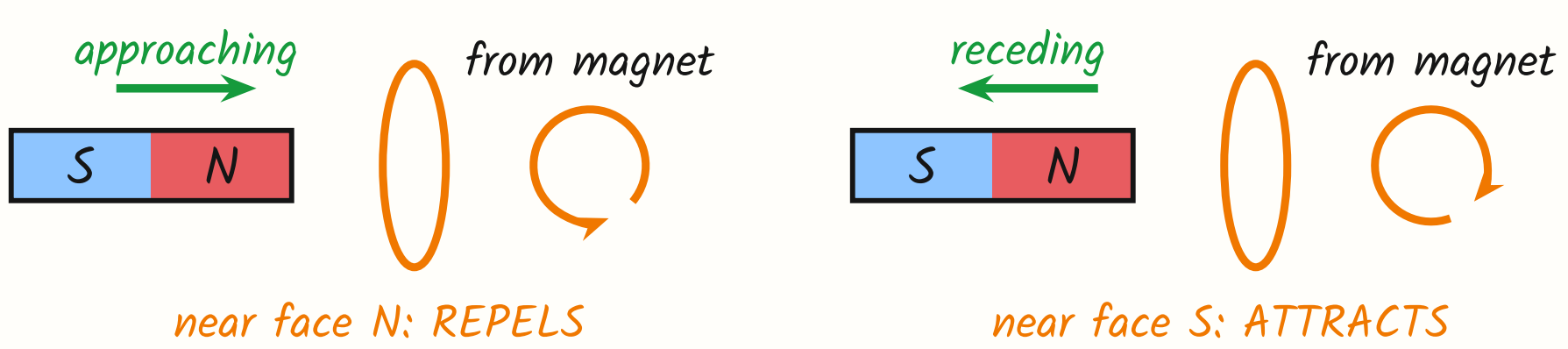
1 Statement

induced current OPPOSES the CHANGE in flux

it opposes the CHANGE, not the flux itself · equivalently it opposes the RELATIVE MOTION that causes it

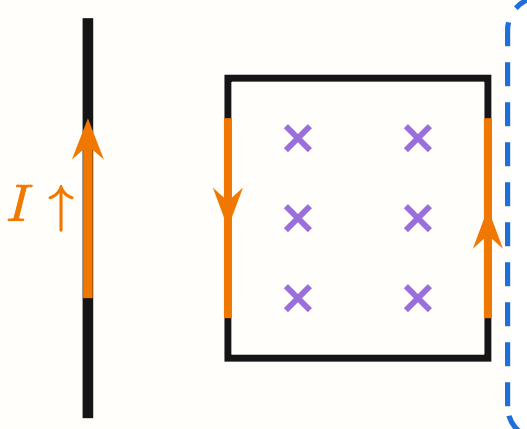
- Energy conservation: if the induced current AIDED the change, the magnet would speed up by itself (energy from nothing). Work done against the opposing force becomes electrical energy, then heat.

2 Magnet and loop (seen from the magnet)



Approaching N pole: current ANTICLOCKWISE seen from the magnet (that face acts as N). Receding: CLOCKWISE (face S).

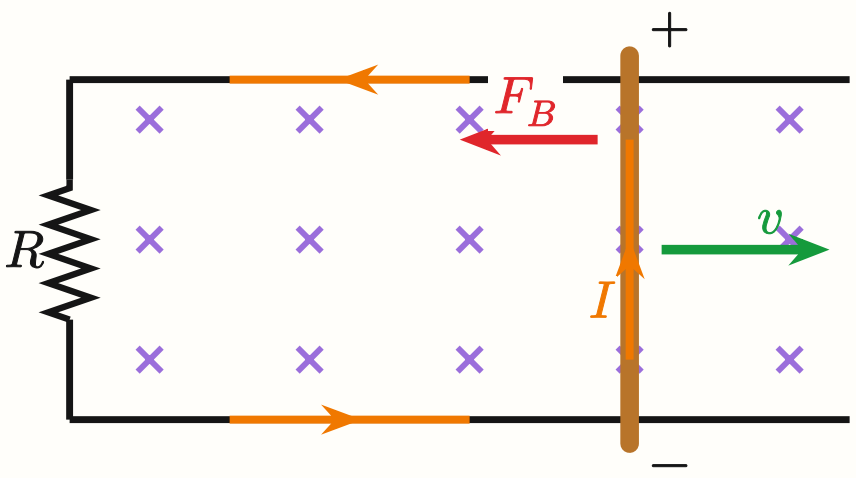
3 Loop beside a long wire



- Long wire, I upward and INCREASING, coplanar loop on its right: flux INTO the page grows.
- Induced current ANTICLOCKWISE; its near side is opposite to I: loop REPELLED.

Motional EMF

1 Rod sliding on rails



$$\varepsilon = Bvl$$

UNIFORM $\vec{B}$; rod, $\vec{v}$ and $\vec{B}$ mutually PERPENDICULAR · general: $\varepsilon = \int (\vec{v} \times \vec{B}) \cdot d\vec{l}$

- Charge q in the rod feels $q\vec{v} \times \vec{B}$ along the rod; charges pile up until $qE = qvB$, so $\varepsilon = El = Bvl$. Here $\vec{v} \times \vec{B}$ points UP the rod: TOP end at higher potential.
- Same answer from flux: $\Phi = Blx$, $\left|\frac{d\Phi}{dt}\right| = Bl\frac{dx}{dt} = Blv$.

2 Current, force, power

$$I = \frac{Bvl}{R} \qquad F_B = \frac{B^2 l^2 v}{R}$$

$$P = F_B v = \frac{B^2 l^2 v^2}{R} = I^2 R$$

total circuit resistance R, rails and rod ideal unless included in R · F_B OPPOSES $\vec{v}$ (Lenz) · constant v needs an external force $F = F_B$

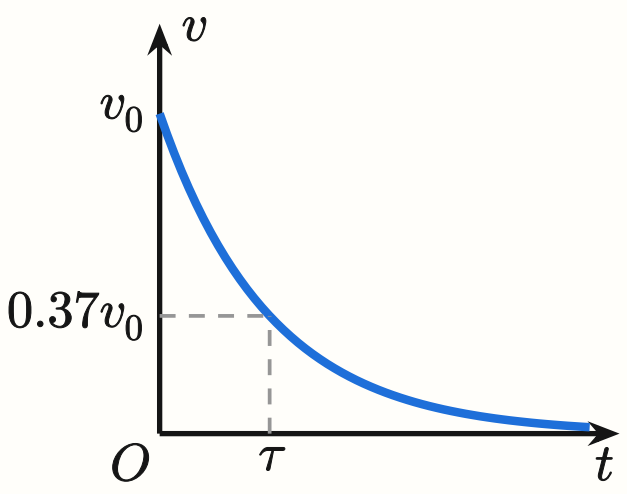
Example

$B = 0.4\,\mathrm{T}$, $l = 0.5\,\mathrm{m}$, $v = 10\,\mathrm{m\,s^{-1}}$, $R = 2\,\Omega$: $\varepsilon = 2\,\mathrm{V}$, $I = 1\,\mathrm{A}$, $F_B = BIl = 0.2\,\mathrm{N}$, $P = Fv = 2\,\mathrm{W} = I^2 R$.

Rod on Rails: Motion

horizontal smooth rails, uniform $\vec{B} \perp$ the plane of the rails, rod of mass m, total resistance R, $\tau = \dfrac{mR}{B^2 l^2}$

1 Rod given speed v_0, then left alone

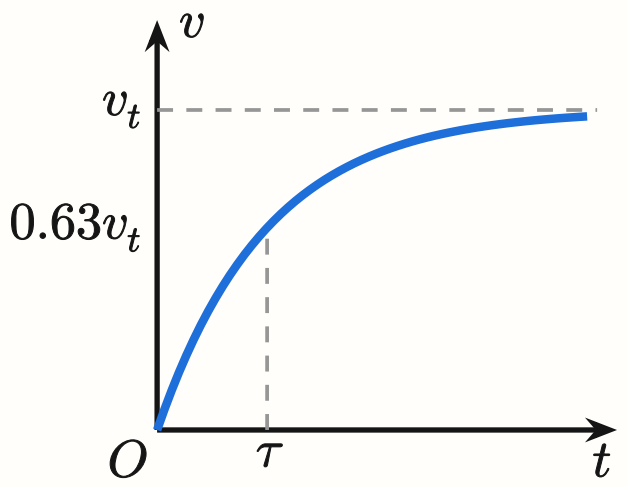


$$v = v_0 e^{-t/\tau}$$

$$x_{stop} = \frac{mv_0 R}{B^2 l^2} = v_0 \tau$$

no external force, no friction · $m\frac{dv}{dt} = -\frac{B^2 l^2 v}{R}$ · heat $= \frac{1}{2}mv_0^2$ · never stops in a finite TIME, yet the DISTANCE is finite

2 Constant force F from rest



$$v = v_t\left(1 - e^{-t/\tau}\right)$$

$$v_t = \frac{FR}{B^2 l^2}$$

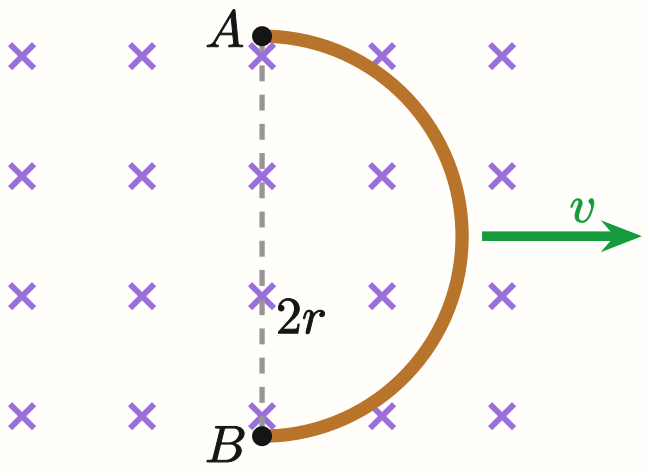
F along the rails, starts from rest · terminal speed when $F = \frac{B^2 l^2 v}{R}$ · then ALL of Fv becomes heat

Example

$m = 0.1\,\text{kg}$, $B = 0.5\,\text{T}$, $l = 0.4\,\text{m}$, $R = 0.2\,\Omega$: $B^2 l^2 = 0.04$, $\tau = \frac{0.1 \times 0.2}{0.04} = 0.5\,\text{s}$. With $F = 0.2\,\text{N}$: $v_t = \frac{0.2 \times 0.2}{0.04} = 1\,\text{m s}^{-1}$. Left alone at $v_0 = 4\,\text{m s}^{-1}$: stops after $v_0\tau = 2\,\text{m}$.

Shaped Conductors, Non-uniform B

1 Arbitrary-shaped conductor

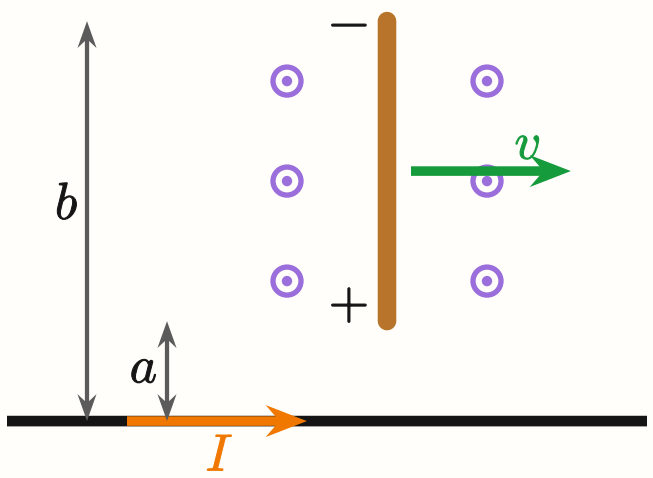


$$\varepsilon = Bv\,L_{eff}$$

UNIFORM $\vec{B}$, conductor in the plane $\perp \vec{B}$, pure translation · L_{eff} = length of the STRAIGHT line joining the ends, measured $\perp \vec{v}$ · semicircle: $\varepsilon = 2rBv$

- *Closed loop TRANSLATING in a uniform field: net $\varepsilon = 0$ (the two halves cancel), although each half has an EMF.*
- *Rod and $\vec{v}$ both $\perp$ uniform $\vec{B}$, $\vec{v}$ at angle θ to the rod: $\varepsilon = Bvl\sin\theta$; $\vec{v}$ along the rod: $\varepsilon = 0$.*

2 Rod in the field of a long wire



$$\varepsilon = \int_a^b \frac{\mu_0 I}{2\pi x} v\,dx = \frac{\mu_0 I v}{2\pi} \ln\frac{b}{a}$$

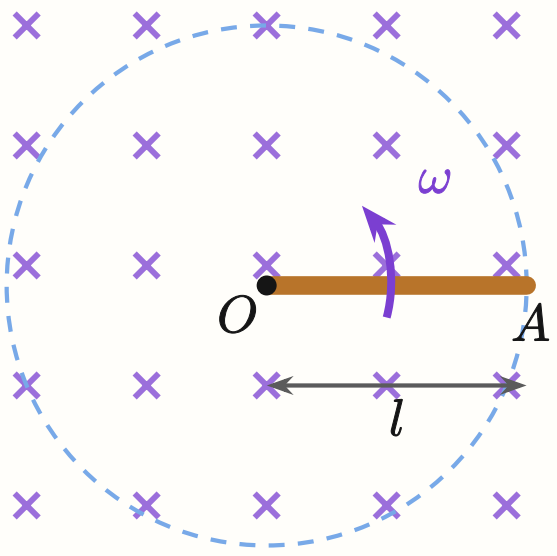
LONG straight wire, rod $\perp$ wire in a plane containing it, $\vec{v} \parallel$ wire · ends at distances a and b

- *$\vec{v}$ along I: $\vec{v} \times \vec{B}$ points TOWARDS the wire, so the end NEARER the wire is at higher potential.*
- *Non-uniform $\vec{B}$: split the rod into dx pieces, $d\varepsilon = B(x)\,v\,dx$, and integrate.*

Rotating Rod and Disc

uniform $\vec{B}$ PARALLEL to the rotation axis, rod $\perp$ axis, angular speed ω constant

1 Rod rotating about one end



$$\varepsilon = \int_0^l B(\omega r)\,dr = \tfrac{1}{2}B\omega l^2$$

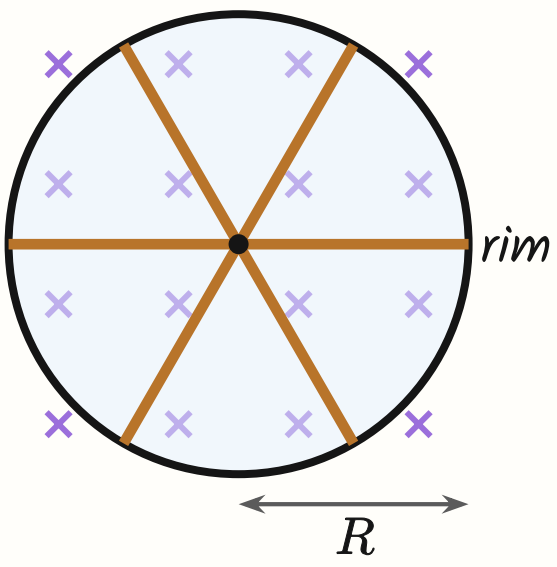
piece dr at radius r moves at $v = \omega r$ · drawn: $\vec{B}$ into the page, ANTICLOCKWISE rotation: $\vec{v} \times \vec{B}$ points to O, so O is at HIGHER potential

- *About its MIDPOINT: the two halves give equal EMFs towards the centre, so V_{ends} difference $= 0$; centre to end $= \tfrac{1}{8}B\omega l^2$.*
- *About a point dividing it into l_1 and l_2: $\varepsilon_{ends} = \tfrac{1}{2}B\omega\left|l_1^2 - l_2^2\right|$.*

Example

$l = 1\,\text{m}$, $\omega = 10\,\text{rad s}^{-1}$, $B = 0.2\,\text{T}$: $\varepsilon = \tfrac{1}{2} \times 0.2 \times 10 \times 1 = 1\,\text{V}$.

2 Disc and wheel with spokes



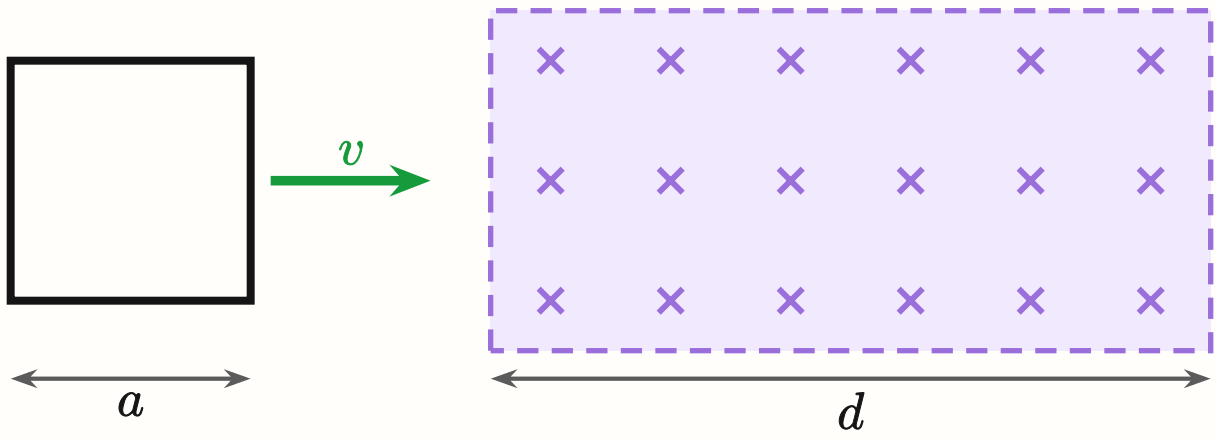
$$\varepsilon_{centre \to rim} = \tfrac{1}{2}B\omega R^2$$

disc (or wheel) of radius R rotating about its axis $\parallel \vec{B}$ · spokes are in PARALLEL: the EMF does not depend on their number

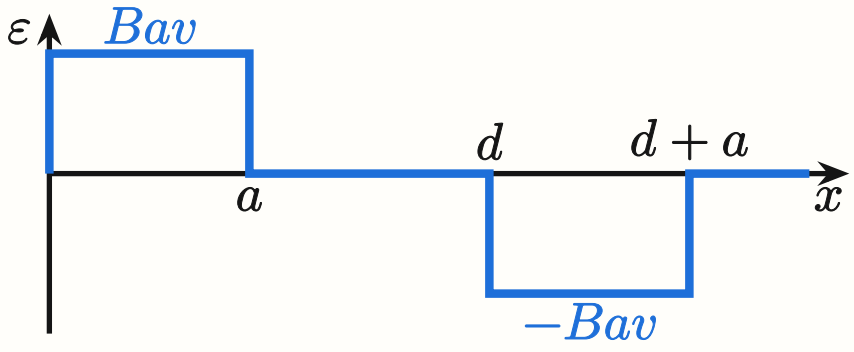
Loop Crossing a Field Region

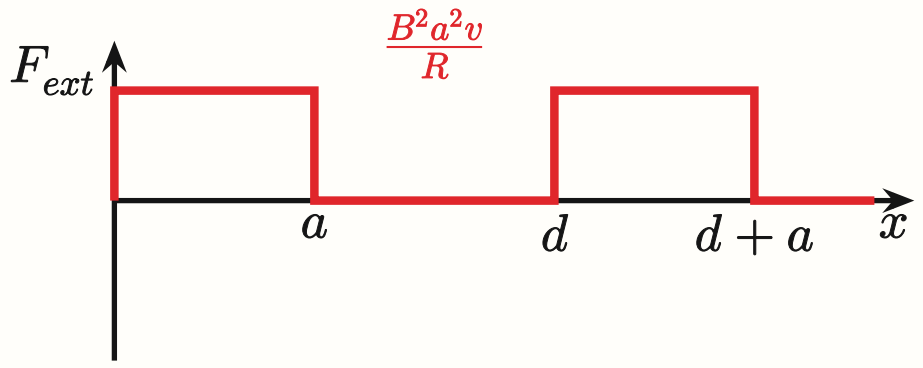
square loop of side a, resistance R, constant speed v, field region of width $d > a$, $\vec{B}$ into the page

1 The set-up



2 EMF and force against x (front edge)





Entering: flux grows, current ANTICLOCKWISE. Fully inside: no current, no force. Leaving: CLOCKWISE. The force needed is ALWAYS along $\vec{v}$.

$$H = \frac{2B^2a^3v}{R}$$

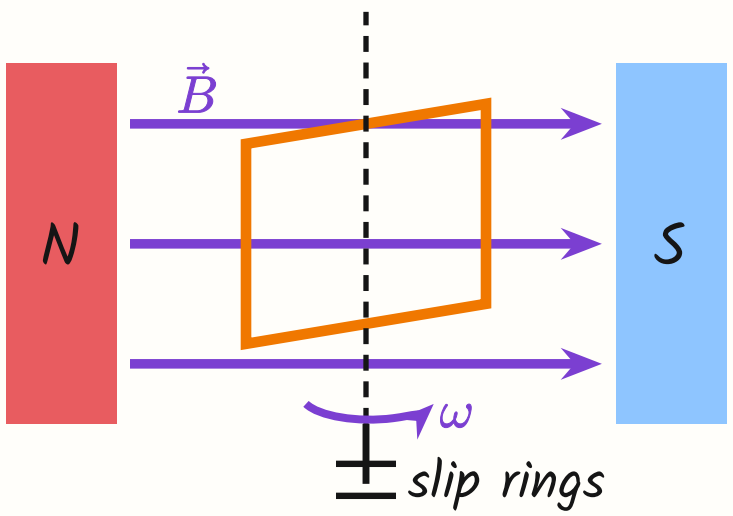
heat for the whole crossing at CONSTANT v, $d > a$ (force $\frac{B^2a^2v}{R}$ over a distance $2a$)

Example

$a = 0.1\,\mathrm{m}$, $B = 0.5\,\mathrm{T}$, $v = 2\,\mathrm{m\,s^{-1}}$, $R = 0.5\,\Omega$: $\varepsilon = 0.1\,\mathrm{V}$, $I = 0.2\,\mathrm{A}$, $F = BIa = 0.01\,\mathrm{N}$, $H = 2Fa = 0.002\,\mathrm{J}$.

AC Generator Principle

1 Coil rotating in a uniform field



$$\Phi = BA\cos\omega t \quad \text{(one turn)}$$

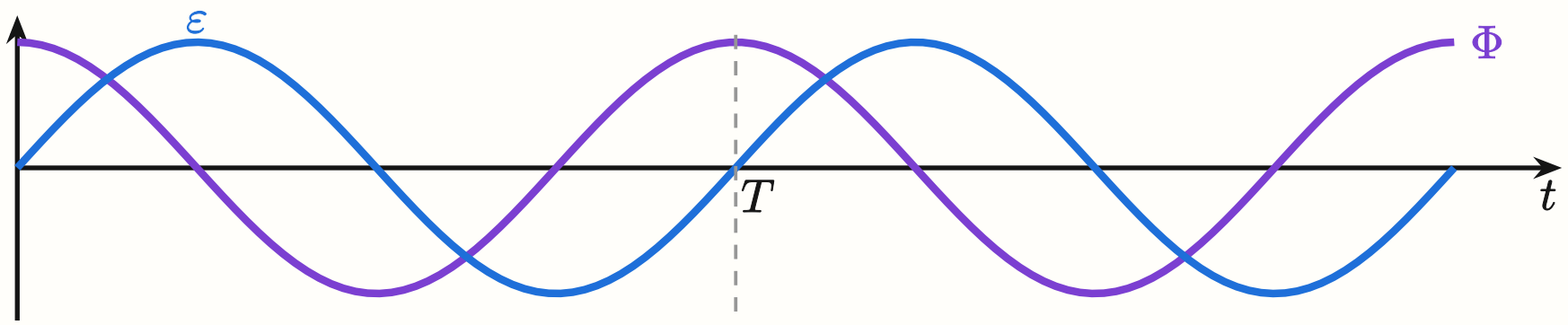
$$\varepsilon = -N\frac{d\Phi}{dt} = NBA\omega\sin\omega t$$

$$\varepsilon_0 = NBA\omega$$

N turns, area A, axis $\perp$ uniform $\vec{B}$, constant ω. $t = 0$: coil plane $\perp \vec{B}$

- ε MAXIMUM when the plane is PARALLEL to $\vec{B}$ ($\Phi = 0$), ZERO when it is $\perp \vec{B}$ (Φ maximum).
- Mechanical work $\rightarrow$ electrical energy: the induced current's torque OPPOSES the rotation.

2 Flux and EMF against time



$\varepsilon = -\frac{d\Phi}{dt}$ LAGS Φ by a quarter period; $T = 2\pi/\omega$.

Example

$N = 100$, $A = 0.05\,\mathrm{m}^2$, $B = 0.4\,\mathrm{T}$, 50 revolutions per second ($\omega = 100\pi = 314.16\,\mathrm{rad\,s^{-1}}$):

$\varepsilon_0 = 100 \times 0.05 \times 0.4 \times 314.16 = 628.3\,\mathrm{V}$.

Induced Electric Field

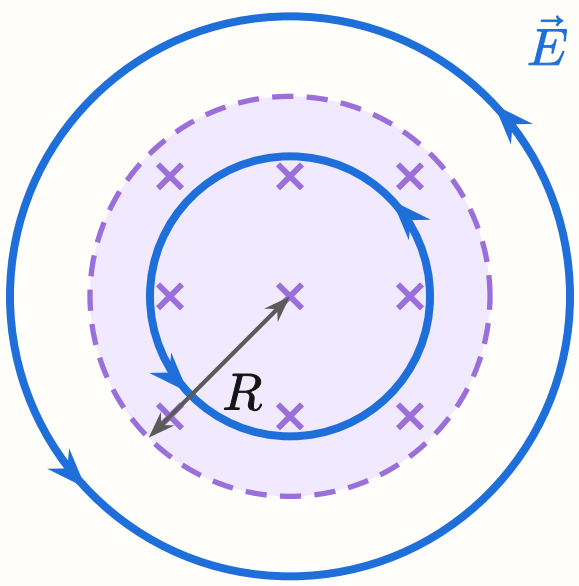
1 A changing $\vec{B}$ makes an $\vec{E}$

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

ANY closed path, even in empty space (no conductor needed) · path direction and $d\vec{A}$ by the right-hand rule

	Electrostatic $\vec{E}$	Induced $\vec{E}$
source	charges	changing $\vec{B}$
field lines	start / end on charges	CLOSED loops
$\oint \vec{E} \cdot d\vec{l}$	0 (conservative)	$\neq 0$ (NON-conservative)
potential	defined	NOT defined

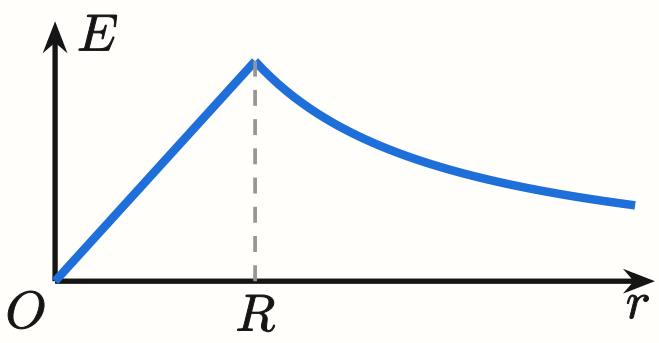
2 Cylindrical field region



$$E = \frac{r\,dB}{2\,dt} \quad (r \leq R)$$

$$E = \frac{R^2\,dB}{2r\,dt} \quad (r \geq R)$$

UNIFORM $B(t)$ in a cylinder of radius R, zero outside · inside: $E \cdot 2\pi r = \pi r^2 \frac{dB}{dt}$ · B into the page INCREASING: $\vec{E}$ ANTICLOCKWISE

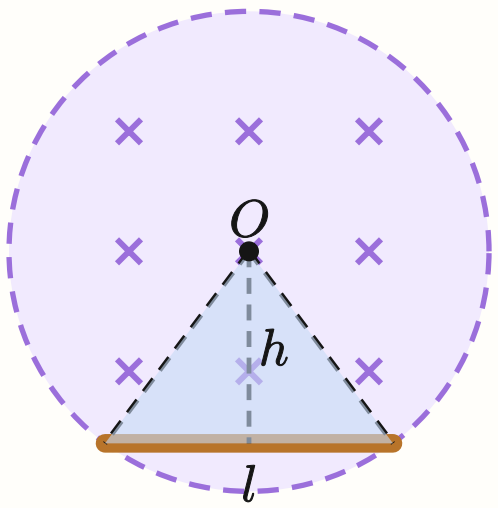


- $E \propto r$ inside, $E \propto \frac{1}{r}$ outside, MAXIMUM at $r = R$.

Induced Field: Special Cases

cylindrical region of radius R, uniform $B(t)$ inside, $\frac{dB}{dt}$ constant

1 Rod as a chord



$$\varepsilon = \frac{dB}{dt} \times \text{area } \triangle = \frac{dB}{dt} \cdot \frac{l}{2}\sqrt{R^2 - \frac{l^2}{4}}$$

rod of length l INSIDE the region, h = distance from the centre O · radial pieces OP, OQ get no EMF ($\vec{E} \perp$ radius)

- *Rod along a DIAMETER: $\varepsilon = 0$. A chord outside the region: $\varepsilon = \frac{dB}{dt} \times$ (field area inside the triangle).*

Example

$\frac{dB}{dt} = 0.02\,\mathrm{T\,s^{-1}}$, $R = 0.1\,\mathrm{m}$, $l = 0.1\,\mathrm{m}$:
$h = \sqrt{0.01 - 0.0025} = 0.0866\,\mathrm{m}$, area
$= \frac{1}{2} \times 0.1 \times 0.0866 = 0.00433\,\mathrm{m^2}$, $\varepsilon = 8.66 \times 10^{-5}\,\mathrm{V}$.

2 Loops

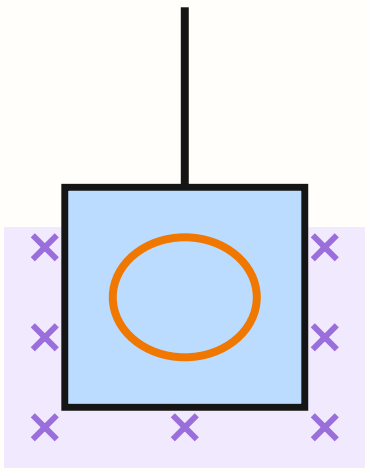
- *Loop enclosing the WHOLE region: $|\varepsilon| = \pi R^2 \frac{dB}{dt}$, whatever its shape and size.*
- *Loop OUTSIDE the region (not enclosing it): $\varepsilon = 0$, though $\vec{E} \neq 0$ on it.*

Eddy Currents

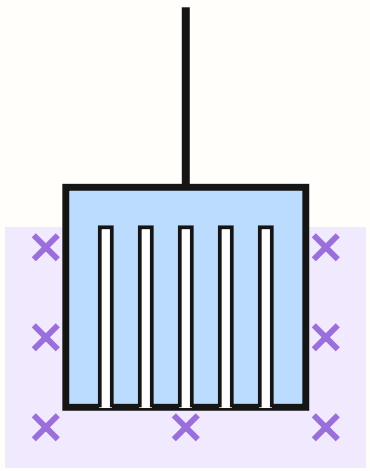
1 What they are

- Currents induced in the BULK of a conductor when the flux through it changes (moving metal in a field, or a changing field). They circulate like eddies and, by Lenz's law, oppose the change.
- They produce heat (I^2R) and a force opposing the motion (magnetic damping).

2 Swinging plate between magnet poles



SOLID plate: large eddies, stops QUICKLY



SLOTTED plate: small eddies, swings LONGER

Golden points

★ Magnet dropped through a vertical copper or aluminium TUBE: eddy currents oppose its fall, it soon falls at a small constant (terminal) speed; a slit tube or a plastic tube: nearly free fall.

Eddy Currents: Uses and Losses

1 Applications (useful eddy currents)

- Electromagnetic BRAKES in trains; magnetic DAMPING: the coil of a moving-coil galvanometer wound on a metal frame comes to rest quickly (dead-beat).
- Induction FURNACE (melting metals with a high-frequency field); induction cooktop; eddy-current speedometers and energy meters.

2 Reducing them (unwanted heat)

- LAMINATED iron cores in transformers and motors: thin sheets insulated from each other, the sheets lying along $\vec{B}$, so the eddy paths are cut; slots in a moving plate do the same.

$\vec{B}$ into the page, changing



SOLID core: large eddies

LAMINATED: sheets along $\vec{B}$, small eddies

Eddy loops lie in planes $\perp \vec{B}$ (here the page); insulated sheets parallel to $\vec{B}$ cut them into thin loops of high resistance: much less heat.

Self-Inductance

L is a property of the coil's GEOMETRY (and core), constant without ferromagnetic material

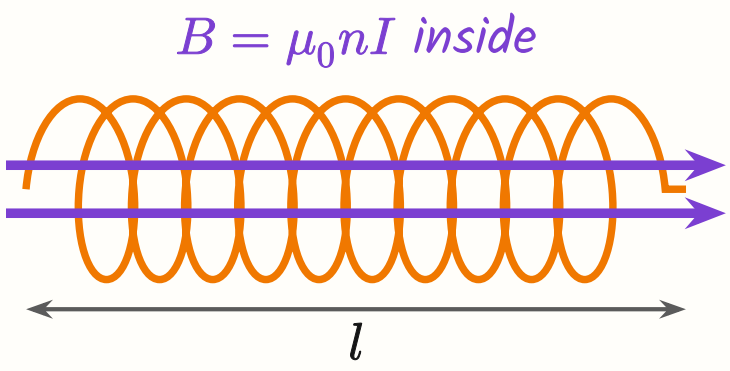
1 Definition

$$N\Phi = LI \qquad \varepsilon = -L\frac{dI}{dt}$$

$N\Phi$ = flux linkage due to the coil's OWN current · unit henry: $1\,\mathrm{H} = 1\,\mathrm{Wb\,A^{-1}} = 1\,\mathrm{V\,s\,A^{-1}}$ · EMF opposes the CHANGE of current (back EMF)

- An inductor is "electrical inertia": it resists a CHANGE of current, not the current itself.

2 Long solenoid



$$L = \mu_0 n^2 A l = \frac{\mu_0 N^2 A}{l}$$

LONG ($l \gg$ radius), air core, end effects ignored · $n = N/l$ turns per metre, A = cross-section · core of relative permeability μ_r: $L \to \mu_r L$

Example

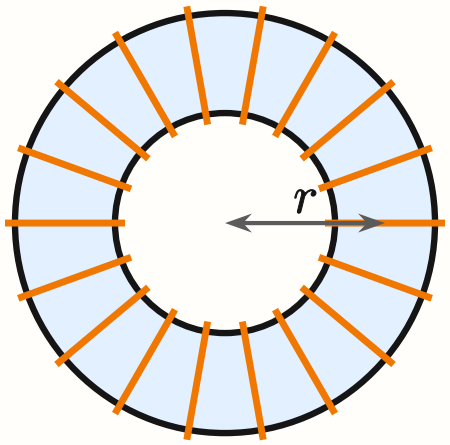
$N = 1000,\ l = 0.5\,\mathrm{m},\ A = 10\,\mathrm{cm^2} = 10^{-3}\,\mathrm{m^2}$:

$L = \frac{4\pi\times10^{-7}\times10^{6}\times10^{-3}}{0.5} = 2.51\times10^{-3}\,\mathrm{H} = 2.51\,\mathrm{mH}.$

Toroid and Back EMF

ideal inductor (L constant, no resistance); sign: ε positive when it drives current in the $+I$ direction

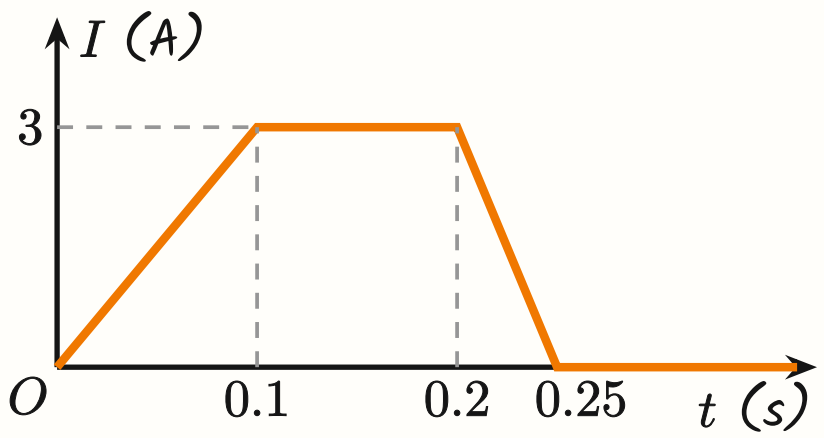
1 Toroid

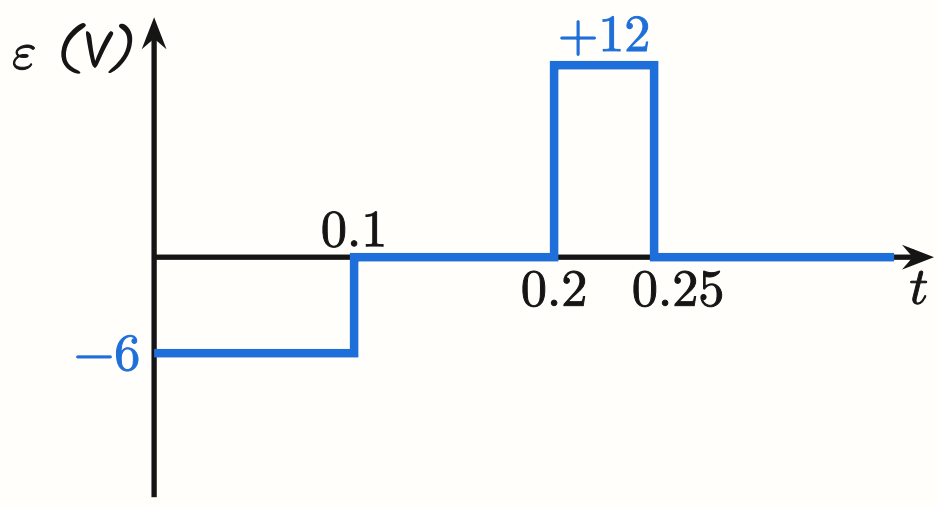


$$L = \frac{\mu_0 N^2 A}{2\pi r}$$

THIN toroid, mean radius $r \gg$ size of the cross-section A · rectangular section (inner a, outer b, height h): $L = \frac{\mu_0 N^2 h}{2\pi} \ln \frac{b}{a}$

2 Current graph → EMF graph





$$\varepsilon = -L\frac{dI}{dt}$$

$L = 0.2\,\text{H}$: *rise* $3\,\text{A}$ *in* $0.1\,\text{s}$ *gives* $\varepsilon = -0.2 \times 30 = -6\,\text{V}$ *(opposes the rise)* · *steady current:* $\varepsilon = 0$ · *fall in* $0.05\,\text{s}$*:* $\varepsilon = +0.2 \times 60 = +12\,\text{V}$ *(tries to keep I going)*

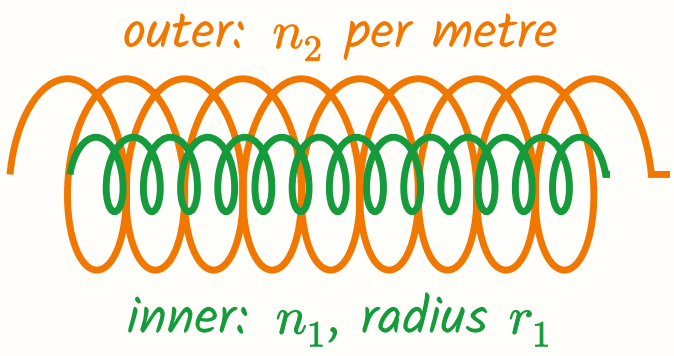
Mutual Inductance

1 Definition

$$N_2\Phi_2 = MI_1 \qquad \varepsilon_2 = -M\frac{dI_1}{dt}$$

Φ_2 = flux through EACH turn of coil 2 due to I_1 · henry · $M_{12} = M_{21}$ · depends on geometry, separation, ORIENTATION, core

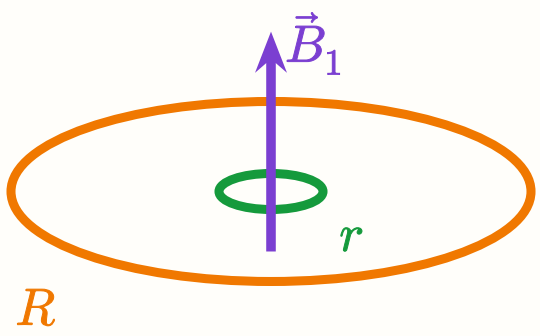
2 Coaxial solenoids



$$M = \mu_0 n_1 n_2 \pi r_1^2 l$$

LONG coaxial solenoids, common length l, inner radius r_1 (the smaller), air core · flux of the OUTER field through the INNER area

3 Small loop at the centre of a large loop



$$M = \frac{\mu_0 \pi r^2}{2R}$$

COAXIAL and COPLANAR, $r \ll R$ · field of the big loop at its centre $\frac{\mu_0 I}{2R}$, uniform over the small one

4 Coefficient of coupling

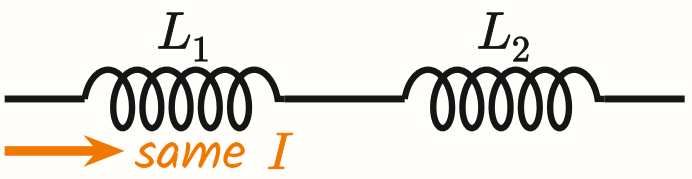
$$k = \frac{M}{\sqrt{L_1 L_2}} \qquad 0 \le k \le 1$$

$k = 1$: all flux of one coil links the other (tightly wound together) · coils with axes PERPENDICULAR: $k \approx 0$

Inductors in Series and Parallel

ideal inductors (no resistance), M = mutual inductance between the two

1 Series

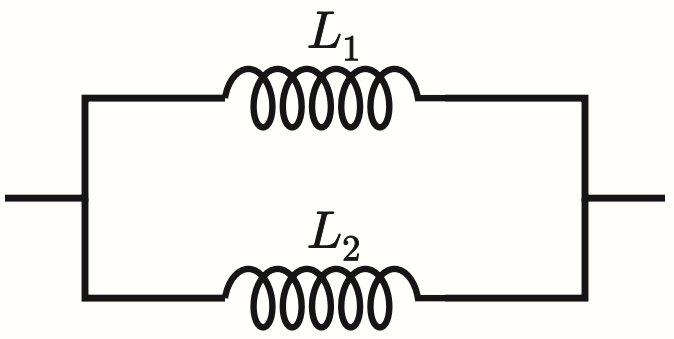


$$L = L_1 + L_2 \pm 2M$$

SAME current through both · +: fluxes AID, −: OPPOSE · no coupling ($M = 0$):

$$L = L_1 + L_2$$

2 Parallel



$$\frac{1}{L} = \frac{1}{L_1} + \frac{1}{L_2}$$

$$L = \frac{L_1 L_2 - M^2}{L_1 + L_2 \mp 2M}$$

SAME voltage across both · first form: NO coupling · second: coupled, − sign when the fluxes AID

Example

$L_1 = 2\,\mathrm{H}$, $L_2 = 8\,\mathrm{H}$: *uncoupled series* $10\,\mathrm{H}$, *parallel* $\frac{2\times 8}{10} = 1.6\,\mathrm{H}$.
With $k = 0.5$: $M = 0.5\sqrt{16} = 2\,\mathrm{H}$, *series aiding* $14\,\mathrm{H}$, *opposing* $6\,\mathrm{H}$.

Golden points

★ *Series opposing:* $L_1 + L_2 - 2M \geq 0$ *always, because* $M \leq \sqrt{L_1 L_2} \leq \frac{L_1 + L_2}{2}$.

Energy in an Inductor

1 Energy stored

$$U = \int_0^I L I'\, dI' = \tfrac{1}{2} L I^2$$

ideal inductor, L constant · work done by the source AGAINST the back EMF, power $P = LI\frac{dI}{dt}$ · stored in the MAGNETIC FIELD

Example

$L = 0.2\,\mathrm{H}$, $I = 3\,\mathrm{A}$: $U = \frac{1}{2} \times 0.2 \times 9 = 0.9\,\mathrm{J}$.

2 Energy density

$$u = \frac{B^2}{2\mu_0}$$

magnetic energy per unit VOLUME ($\mathrm{J\,m^{-3}}$) in vacuum or air · in a linear medium $\frac{B^2}{2\mu}$ · any field shape: $U = \int u\, dV$

- *Check, long solenoid:* $\frac{(\mu_0 n I)^2}{2\mu_0} \times Al = \frac{1}{2}(\mu_0 n^2 Al) I^2 = \frac{1}{2} L I^2$.
- $B = 1\,\mathrm{T}$: $u = \frac{1}{2 \times 4\pi \times 10^{-7}} = 3.98 \times 10^5\,\mathrm{J\,m^{-3}}$.

3 Two coupled coils

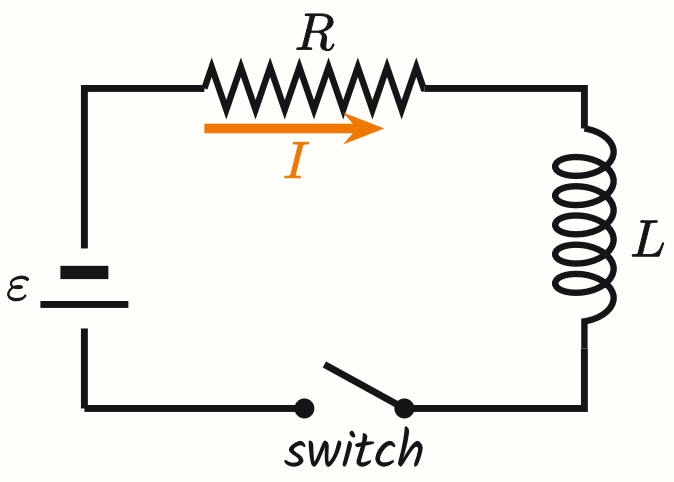
$$U = \tfrac{1}{2} L_1 I_1^2 + \tfrac{1}{2} L_2 I_2^2 \pm M I_1 I_2$$

+ when the fluxes AID, − when they OPPOSE · $U \geq 0$ needs $M \leq \sqrt{L_1 L_2}$

LR Circuit: Growth

ideal battery ε, ideal inductor, R = total resistance, switch CLOSED at $t = 0$, $I(0) = 0$

1 Circuit and equation



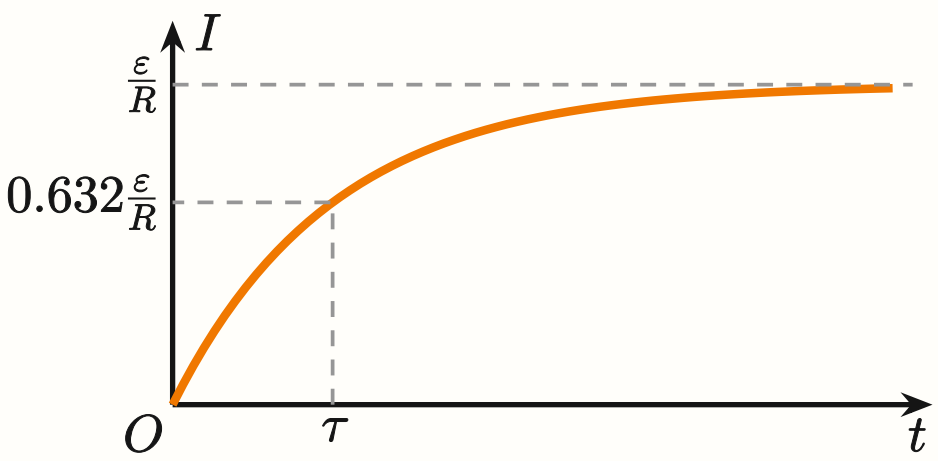
$$\varepsilon - IR - L\frac{dI}{dt} = 0$$

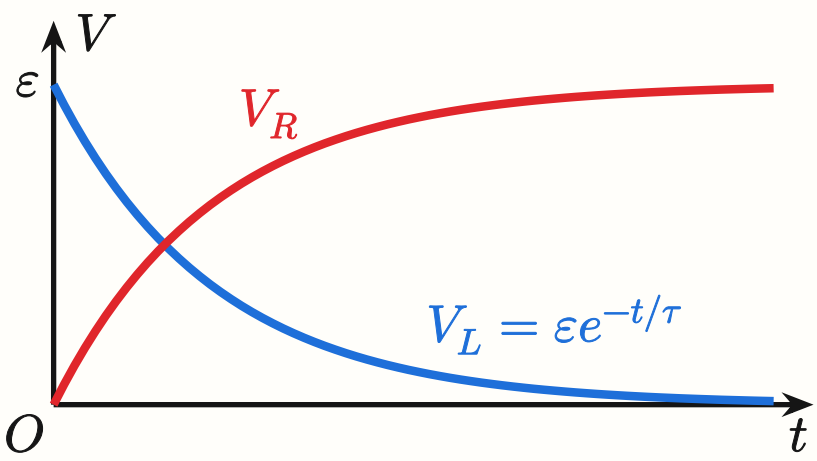
$$I = \frac{\varepsilon}{R}\left(1 - e^{-t/\tau}\right)$$

$$\tau = \frac{L}{R}$$

τ in s · $I_0 = \varepsilon/R$ = STEADY current

2 Graphs





- $t = 0$: inductor = OPEN switch, $V_L = \varepsilon$, $\frac{dI}{dt} = \frac{\varepsilon}{L}$. $t \gg \tau$: plain wire. I/I_0: 63.2% at τ, 86.5% at 2τ, 99.3% at 5τ.

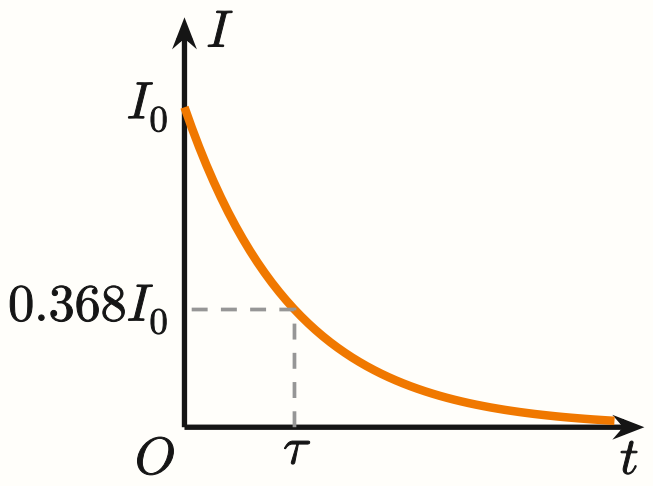
Example

$\varepsilon = 10\,\text{V}$, $R = 5\,\Omega$, $L = 0.1\,\text{H}$: $\tau = 0.02\,\text{s}$, $I_0 = 2\,\text{A}$, at $t = 0.02\,\text{s}$: $I = 2(1 - e^{-1}) = 1.26\,\text{A}$.

LR Circuit: Decay

ideal inductor carrying I_0, battery removed and the inductor shorted through R at $t = 0$

1 Decay of current



$$I = I_0 e^{-t/\tau}$$

$$\tau = \frac{L}{R}$$

R = TOTAL resistance of the loop the inductor now drives · I keeps its DIRECTION while it decays

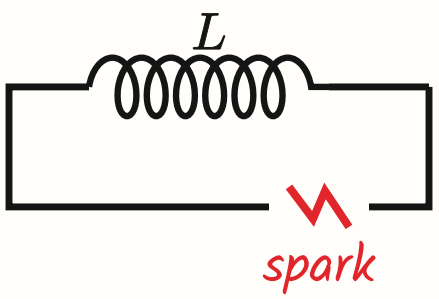
$$t_{1/2} = \tau \ln 2 \qquad H = \tfrac{1}{2} L I_0^2$$

time for the current to halve · total heat in R = energy that was stored

Example

$L = 0.5\,\text{H},\ R = 10\,\Omega,\ I_0 = 2\,\text{A}$: $\tau = 0.05\,\text{s}$,
$t_{1/2} = 0.05 \times 0.693 = 0.0347\,\text{s},\ H = \tfrac{1}{2} \times 0.5 \times 4 = 1\,\text{J}.$

2 Breaking a circuit suddenly



- Opening a switch with NO other path makes dI/dt huge: $|\varepsilon| = L\left|\frac{dI}{dt}\right|$ can be thousands of volts, and the air gap sparks.
- Remedy: a resistor or diode across the coil gives the current a path to decay.

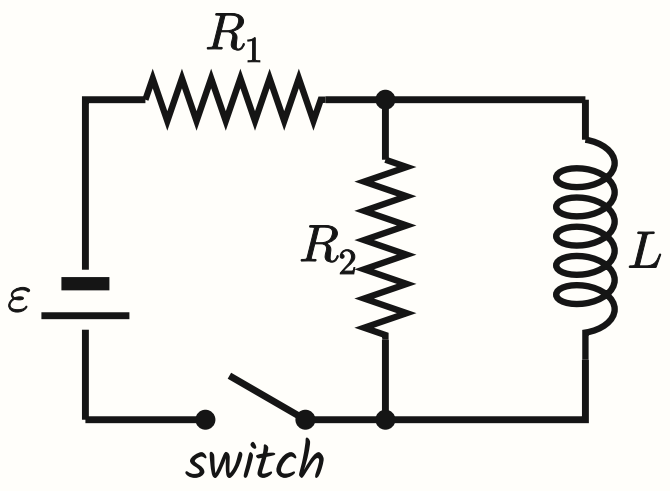
LR Circuits: Quick Analysis

ideal inductors, ideal battery; "long time" = $t \gg \tau$

1 Three rules

- Current through an inductor is CONTINUOUS: $I_L(0^+) = I_L(0^-)$.
- Just after switching on (inductor had no current): treat it as an OPEN circuit.
- Long after: $\frac{dI}{dt} = 0$, an ideal inductor is a SHORT (plain wire).
- Time constant $\tau = L/R_{eq}$, R_{eq} = resistance seen by the inductor with the battery replaced by a wire.

2 Worked circuit



$$I(0^+) = \frac{\varepsilon}{R_1 + R_2}$$

$$I(\infty) = \frac{\varepsilon}{R_1}$$

$$\tau = \frac{L}{R_1 \parallel R_2}$$

battery current, switch closed at $t = 0$ · $R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2}$

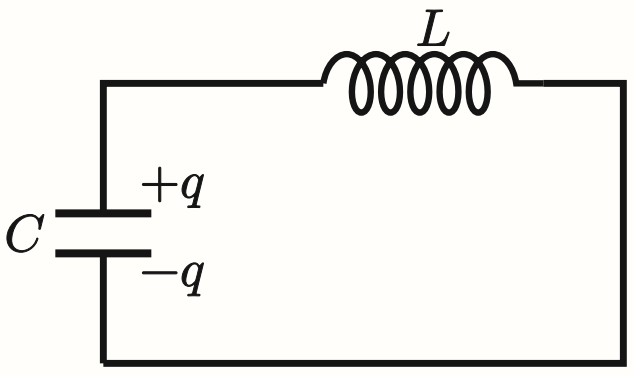
Example

$\varepsilon = 12\,\text{V}$, $R_1 = R_2 = 4\,\Omega$, $L = 2\,\text{H}$: $I(0^+) = \frac{12}{8} = 1.5\,\text{A}$, $I(\infty) = 3\,\text{A}$ (all through L), $\tau = \frac{2}{2} = 1\,\text{s}$, $I_L = 3\,(1 - e^{-t})$ A with t in s.

LC Oscillations

ideal L and C (no resistance); capacitor charged to q_0, connected to L at $t = 0$

1 Charge and current



$$L\frac{d^2q}{dt^2} + \frac{q}{C} = 0$$

$$q = q_0 \cos \omega t, \quad \omega = \frac{1}{\sqrt{LC}}$$

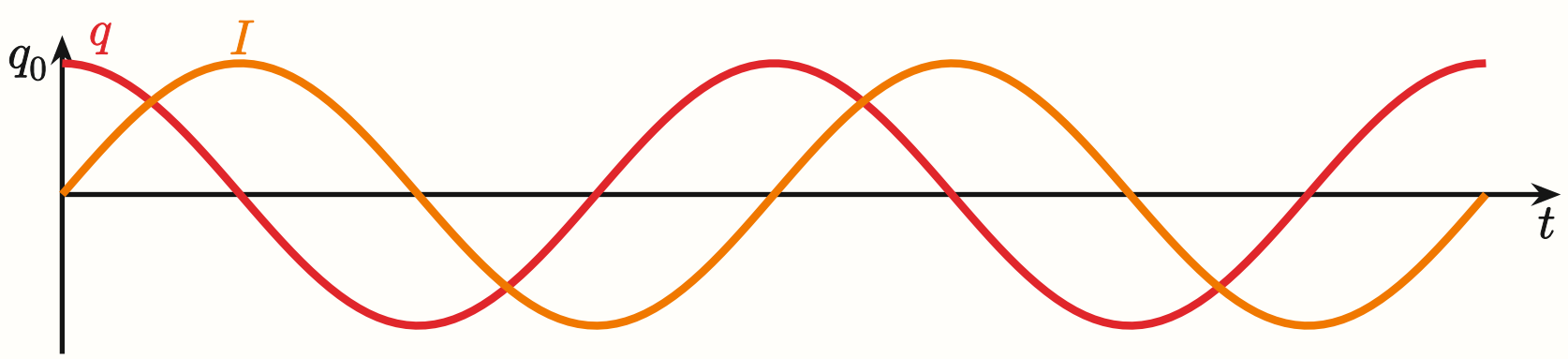
$$I = -\frac{dq}{dt} = I_0 \sin \omega t, \quad I_0 = \omega q_0$$

I = discharge current · $f = \dfrac{1}{2\pi\sqrt{LC}}$

2 Spring analogy

- $q \leftrightarrow x$, $I \leftrightarrow v$, $L \leftrightarrow m$, $\frac{1}{C} \leftrightarrow k$: $\omega = \frac{1}{\sqrt{LC}}$ like $\sqrt{k/m}$. With resistance (LCR) the oscillation decays.

3 q and I against time

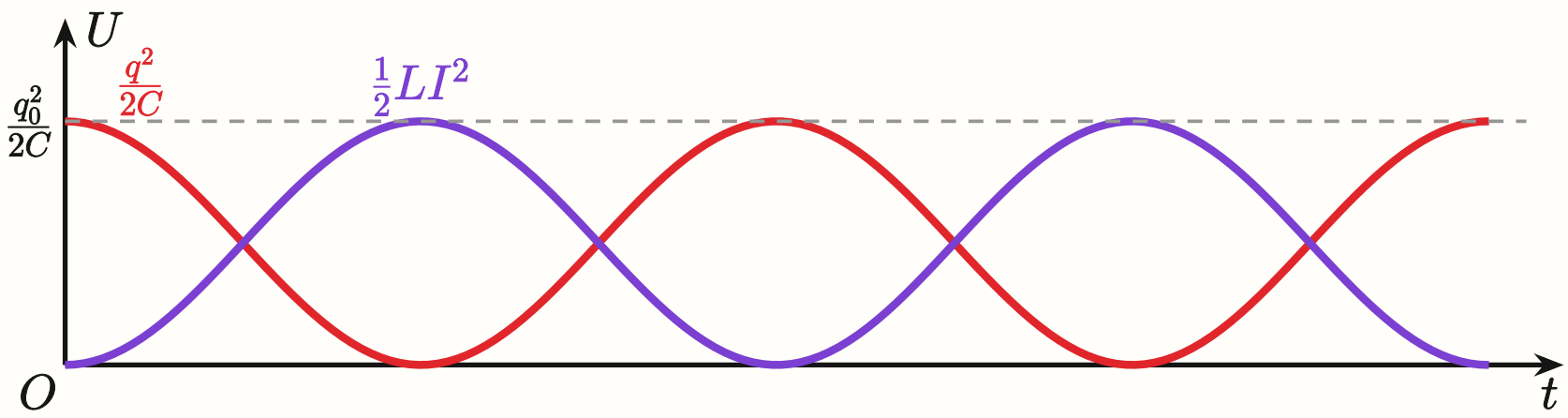


I is maximum when $q = 0$ and zero when $q = \pm q_0$: a quarter period apart.

LC Oscillations: Energy

ideal L and C (no resistance), no radiation

1 Energy swings between C and L



$$\frac{q^2}{2C}+\frac{1}{2}LI^2=\frac{q_0^2}{2C}$$

total energy CONSTANT (no resistance, no radiation) · $U_C=\frac{q_0^2}{2C}\cos^2\omega t$, $U_L=\frac{q_0^2}{2C}\sin^2\omega t$

Example

$L=10\,\text{mH}$, $C=100\,\mu\text{F}$, $q_0=1\,\text{mC}$: $\omega=1000\,\text{rad}\,\text{s}^{-1}$, $f=159.2\,\text{Hz}$, $I_0=1\,\text{A}$, *energy* $5\,\text{mJ}$.

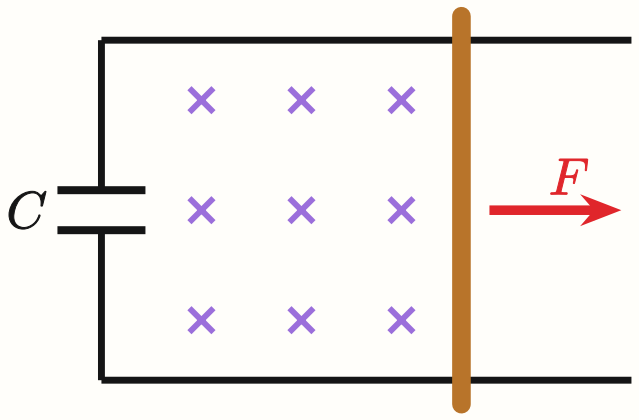
Golden points

★ $|I|=\omega\sqrt{q_0^2-q^2}$, *just like* $v=\omega\sqrt{A^2-x^2}$ *in SHM;* $U_C=U_L$ *when* $q=\frac{q_0}{\sqrt{2}}$.

★ *Each energy repeats every HALF period: its frequency is* $2f$.

Special Cases (I)

1 Rod on rails with a capacitor

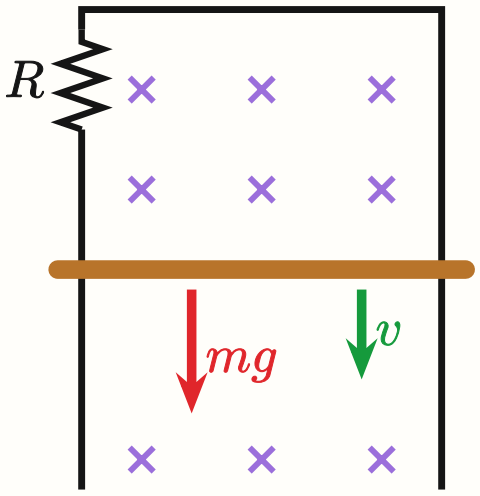


$$a = \frac{F}{m + CB^2l^2}$$

constant force F, no resistance, smooth rails, uniform $\vec{B} \perp$ the plane of the rails · $q = CBlv$, $I = CBla$ CONSTANT · UNIFORM acceleration

---

2 Rod falling on vertical rails



$$v_t = \frac{mgR}{B^2l^2}$$

rails vertical, uniform $\vec{B}$ horizontal and $\perp$ the plane of the rails, total resistance R, no friction · terminal when $mg = \frac{B^2l^2v}{R}$ · then power $mgv_t = I^2R$

---

3 Ring with a cut

- *A ring with a narrow GAP in a changing flux: full EMF $\varepsilon = -\frac{d\Phi}{dt}$ across the gap but NO current, so no heat and no force.*
- *Magnet dropped through a CLOSED ring: $a < g$ both while entering and while leaving; through a CUT ring: $a = g$ (air drag neglected).*

Special Cases (2)

1 Charge and flux

- Charge through a coil depends only on $\Delta\Phi$: pulling a magnet out fast or slow sends the SAME charge $N\Delta\Phi/R$; the heat is LARGER when it is fast.
- Coil flipped through 180° in a uniform $\vec{B}$ (plane $\perp \vec{B}$ at the start): $\Delta\Phi = 2BA$ per turn.

Example

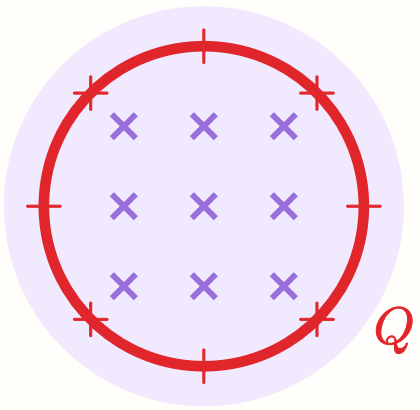
Coil of 50 turns, area $0.01\,\mathrm{m}^2$, total resistance $5\,\Omega$, plane $\perp$ uniform $B = 0.4\,\mathrm{T}$, pulled right out of the field:
$q = \frac{50\times0.4\times0.01}{5} = 0.04\,\mathrm{C}$, however fast it is pulled.

Golden points

★ Two conductors with EQUAL EMFs in a loop (e.g. two rods moving together at the same speed in a uniform $\vec{B}$): net EMF zero, no current.

★ Self-induced EMF is zero at the instant I is maximum ($dI/dt = 0$), not at $I = 0$.

2 Charged ring in a changing field



$$\Delta L = \frac{Qr^2}{2}\Delta B$$

non-conducting ring, charge Q, radius r, free to rotate about its axis, inside a cylindrical region of uniform $B(t)$ · torque $QEr = \frac{Qr^2}{2}\frac{dB}{dt}$

Standard Results

1 One-line results

Quantity	Result	Condition
Faraday	$\varepsilon = -N\frac{d\Phi}{dt}$	N turns, same flux each
Charge	$q = \frac{N\Delta\Phi}{R}$	total resistance R
Rod on rails	$\varepsilon = Bvl$	uniform $\vec{B}$; $\vec{B}$, $\vec{v}$, rod mutually $\perp$
Rotating rod	$\frac{1}{2}B\omega l^2$	about one end, axis $\parallel \vec{B}$
Disc	$\frac{1}{2}B\omega R^2$	centre to rim, axis $\parallel \vec{B}$
Generator	$\varepsilon_0 = NBA\omega$	N turns, constant ω, axis $\perp$ uniform $\vec{B}$
Induced E	$\frac{r}{2}\frac{dB}{dt}$, $\frac{R^2}{2r}\frac{dB}{dt}$	uniform $B(t)$ in the cylinder, zero outside: inside, outside
Solenoid	$L = \mu_0 n^2 Al$	long, air core
Coaxial	$M = \mu_0 n_1 n_2 \pi r_1^2 l$	r_1 = inner radius
Coupling	$M = k\sqrt{L_1 L_2}$	$0 \le k \le 1$
Energy	$\frac{1}{2}LI^2$, $u = \frac{B^2}{2\mu_0}$	ideal L; vacuum
LR	$\tau = \frac{L}{R}$	R = total resistance in the loop the inductor drives; growth $1 - e^{-t/\tau}$, decay $e^{-t/\tau}$
LC	$\omega = \frac{1}{\sqrt{LC}}$	no resistance

Summary

$$\varepsilon = -N\frac{d\Phi}{dt} \;\bullet\; \varepsilon = Bvl \;\bullet\; U = \tfrac{1}{2}LI^2 \;\bullet\; \tau = \frac{L}{R}$$

uniform $\vec{B}$, rod $\perp \vec{B}$ and $\vec{v}$ · ideal inductor · R = total resistance

Common Traps

1 Exam traps

Avoid these

- ✗ Big flux = big EMF (no: EMF needs a CHANGING flux; generator: Φ maximum, $\varepsilon = 0$).
- ✗ Lenz: the induced current opposes the FLUX (no: it opposes the CHANGE; decreasing flux is AIDED).
- ✗ θ in $\Phi = BA\cos\theta$ measured from the PLANE (no: from the NORMAL).
- ✗ A loop translating in a UNIFORM field has a current (no: net $\varepsilon = 0$).
- ✗ $\varepsilon = Blv$ for any orientation (only rod, $\vec{v}$, $\vec{B}$ mutually $\perp$; else use $(\vec{v} \times \vec{B}) \cdot \vec{l}$).
- ✗ Induced $\vec{E}$ has a potential (no: NON-conservative; EMF depends on the path).
- ✗ Charge flow depends on the speed of the magnet (no: $q = N\Delta\Phi/R$ only).
- ✗ Inductor current jumps at the switch (no: it is CONTINUOUS; the voltage can jump).
- ✗ $\tau = L/R$ with the WRONG R (use the resistance in the loop the inductor drives, battery shorted).
- ✗ Rotating rod: $\varepsilon = B\omega l^2$ (no: the average speed is $\omega l/2$, so $\frac{1}{2}B\omega l^2$).

Golden points

- ★ First ask: what changes (B, area, angle)? is $\vec{B}$ uniform? what is the TOTAL resistance? which way does Lenz push?

Questions: JEE Main Level (I)

uniform $\vec{B}$, $\mu_0 = 4\pi \times 10^{-7}\,\mathrm{T\,m\,A^{-1}}$, ideal inductors

Q1 · JEE Main level · MCQ

Coil: 50 turns, area $0.02\,\mathrm{m^2}$, plane $\perp$ uniform $\vec{B}$. B rises uniformly from $0.1\,\mathrm{T}$ to $0.5\,\mathrm{T}$ in $0.2\,\mathrm{s}$. Induced EMF: (A) $2\,\mathrm{V}$ (B) $1\,\mathrm{V}$ (C) $4\,\mathrm{V}$ (D) $0.2\,\mathrm{V}$

Solution: $\varepsilon = NA\frac{\Delta B}{\Delta t} = 50 \times 0.02 \times \frac{0.4}{0.2} = 2\,\mathrm{V}.$

Answer: (A)

Q2 · JEE Main level · Numerical

Coil: 100 turns, area $0.01\,\mathrm{m^2}$, total resistance $10\,\Omega$, plane $\perp \vec{B}$ ($B = 0.5\,\mathrm{T}$), turned over through $180°$. Charge (in C) that flows?

Solution: $\Delta\Phi = 2BA = 0.01\,\mathrm{Wb}$ *per turn;*
$q = \frac{N\Delta\Phi}{R} = \frac{100\times0.01}{10} = 0.1\,\mathrm{C}.$

Answer: 0.1

Q3 · JEE Main level · MCQ

A $0.5\,\mathrm{m}$ rod slides at a constant $4\,\mathrm{m\,s^{-1}}$ on ideal rails closed by $2\,\Omega$; uniform $B = 0.2\,\mathrm{T} \perp$ the plane of the rails. Force needed: (A) $0.01\,\mathrm{N}$ (B) $0.02\,\mathrm{N}$ (C) $0.04\,\mathrm{N}$ (D) $0.2\,\mathrm{N}$

Solution: $\varepsilon = Bvl = 0.4\,\mathrm{V}$, $I = 0.2\,\mathrm{A}$,
$F = BIl = 0.2 \times 0.2 \times 0.5 = 0.02\,\mathrm{N}.$

Answer: (B)

Q4 · JEE Main level · Numerical

A $1\,\mathrm{m}$ metal rod rotates at $20\,\mathrm{rad\,s^{-1}}$ about one end, in a plane $\perp$ a uniform $B = 0.5\,\mathrm{T}$. EMF between its ends (in V)?

Solution: $\varepsilon = \frac{1}{2}B\omega l^2 = \frac{1}{2} \times 0.5 \times 20 \times 1 = 5\,\mathrm{V}.$

Answer: 5

Questions: JEE Main Level (2)

$\mu_0 = 4\pi \times 10^{-7}\,\mathrm{T\,m\,A^{-1}}$, *ideal battery,* t *from closing the switch*

Q5 · JEE Main level · MCQ

Long air-cored solenoid: 500 *turns, length* $0.5\,\mathrm{m}$, *area* $4 \times 10^{-4}\,\mathrm{m^2}$. $L \approx$ *(A)* $0.25\,\mathrm{mH}$ *(B)* $2.5\,\mathrm{mH}$ *(C)* $0.5\,\mathrm{mH}$ *(D)* $25\,\mathrm{mH}$

Solution: $L = \frac{\mu_0 N^2 A}{l} = \frac{4\pi\times10^{-7}\times2.5\times10^{5}\times4\times10^{-4}}{0.5} = 2.51 \times 10^{-4}\,\mathrm{H}$.

Answer: (A)

Q6 · JEE Main level · Numerical

Current in a $50\,\mathrm{mH}$ *coil falls uniformly from* $2\,\mathrm{A}$ *to zero in* $0.01\,\mathrm{s}$. *(i) EMF (in V)? (ii) Initial stored energy (in J)?*

Solution: *(i)* $\varepsilon = L\frac{\Delta I}{\Delta t} = 0.05 \times \frac{2}{0.01} = 10\,\mathrm{V}$. *(ii)* $U = \frac{1}{2} \times 0.05 \times 2^2 = 0.1\,\mathrm{J}$.

Answer: (i) 10 **(ii)** 0.1

Q7 · JEE Main level · MCQ

$L = 2\,\mathrm{H}$, $R = 4\,\Omega$, $\varepsilon = 12\,\mathrm{V}$ *in series. Time to reach HALF the steady current:* *(A)* $0.35\,\mathrm{s}$ *(B)* $0.5\,\mathrm{s}$ *(C)* $0.69\,\mathrm{s}$ *(D)* $0.17\,\mathrm{s}$

Solution: $\tau = \frac{L}{R} = 0.5\,\mathrm{s}$; $1 - e^{-t/\tau} = \frac{1}{2}$ *gives* $t = \tau \ln 2 = 0.5 \times 0.693 = 0.35\,\mathrm{s}$.

Answer: (A)

Q8 · JEE Main level · Numerical

$L_1 = 0.1\,\mathrm{H}$, $L_2 = 0.4\,\mathrm{H}$, *coupling coefficient* 0.5; I_1 *changes at* $20\,\mathrm{A\,s^{-1}}$. *EMF in coil 2 (in V)?*

Solution: $M = k\sqrt{L_1 L_2} = 0.5 \times 0.2 = 0.1\,\mathrm{H}$; $\varepsilon_2 = M\frac{dI_1}{dt} = 0.1 \times 20 = 2\,\mathrm{V}$.

Answer: 2

Questions: JEE Advanced Level (I)

uniform $\vec{B}$, smooth rails, ideal rails and rod, R = total resistance

A1 · JEE Advanced level · Multi-correct

A rod of mass $0.2\,\mathrm{kg}$, length $1\,\mathrm{m}$ starts from rest on horizontal rails closed by $R = 1\,\Omega$; uniform $B = 0.5\,\mathrm{T} \perp$ the plane of the rails; a constant force $0.5\,\mathrm{N}$ pulls it along the rails. Then: (A) terminal speed $2\,\mathrm{m\,s^{-1}}$ (B) time constant $0.8\,\mathrm{s}$ (C) at terminal speed all the work of the force becomes heat (D) the current at $t = 0.8\,\mathrm{s}$ is $0.5\,\mathrm{A}$

Solution: $B^2l^2 = 0.25$: $v_t = \frac{FR}{B^2l^2} = 2\,\mathrm{m\,s^{-1}}$, $\tau = \frac{mR}{B^2l^2} = 0.8\,\mathrm{s}$. At v_t: $I = \frac{Blv_t}{R} = 1\,\mathrm{A}$, $I^2R = 1\,\mathrm{W} = Fv_t$. At $t = \tau$: $I = 1 \times (1 - e^{-1}) = 0.632\,\mathrm{A}$.

Answer: (A), (B), (C)

A2 · JEE Advanced level · Numerical

A copper disc of radius $0.2\,\mathrm{m}$ spins at $50\,\mathrm{rad\,s^{-1}}$ about its axis, parallel to $B = 0.4\,\mathrm{T}$. A $2\,\Omega$ resistor joins the axle to the rim (disc resistance negligible). (i) Current (in A)? (ii) Torque (in N m) needed to keep ω constant?

Solution: $\varepsilon = \frac{1}{2}B\omega R^2 = \frac{1}{2} \times 0.4 \times 50 \times 0.04 = 0.4\,\mathrm{V}$, $I = 0.2\,\mathrm{A}$. Torque of the magnetic force on the radial current: $\int_0^R IBr\,dr = \frac{IBR^2}{2} = \frac{0.2 \times 0.4 \times 0.04}{2} = 0.0016\,\mathrm{N\,m}$. Check: torque $\times\,\omega = 0.08\,\mathrm{W} = I^2R$.

Answer: (i) 0.2 (ii) 0.0016

Golden points

- ★ Power check in every motional-EMF problem: mechanical power in $= I^2R$ (at constant speed).

Questions: JEE Advanced Level (2)

ideal battery and inductor; cylindrical field region radius R, $\frac{dB}{dt} = \beta > 0$ constant

A3 · JEE Advanced level · Multi-correct

Uniform $B(t)$ inside a long cylinder of radius R, zero outside, $\frac{dB}{dt} = \beta$. Which are correct? (A) $E = \frac{\beta r}{2}$ for $r < R$ (B) E is maximum at $r = R$ (C) $\oint \vec{E} \cdot d\vec{l}$ round a circle of radius $2R$ is $4\pi R^2 \beta$ in magnitude (D) a positive charge q taken once round, along the induced $\vec{E}$, on a circle of radius $r > R$ gains energy $q\pi R^2 \beta$

Solution: Inside: $E \cdot 2\pi r = \pi r^2 \beta$. Outside: $E = \frac{R^2 \beta}{2r}$, so E peaks at $r = R$. Flux only inside R: $\oint = \pi R^2 \beta$, not $4\pi R^2 \beta$; work per round $= q\pi R^2 \beta$ for every $r > R$.

Answer: (A), (B), (D)

A4 · JEE Advanced level · Numerical

A $12\,\mathrm{V}$ battery feeds two parallel branches: $R_1 = 6\,\Omega$ alone, and $L = 3\,\mathrm{H}$ in series with $R_2 = 3\,\Omega$. (i) Battery current just after closing (in A)? (ii) Long after? (iii) Then the battery is disconnected: heat (in J) produced in R_1?

Solution: (i) $I_L = 0$: $\frac{12}{6} = 2\,\mathrm{A}$. (ii) $2 + \frac{12}{3} = 6\,\mathrm{A}$. (iii) $I_L = 4\,\mathrm{A}$ decays through $R_1 + R_2$: total heat $\frac{1}{2} \times 3 \times 16 = 24\,\mathrm{J}$, shared in the ratio of resistances (same current): R_1 gets $\frac{6}{9} \times 24 = 16\,\mathrm{J}$.

Answer: (i) 2 (ii) 6 (iii) 16

Golden points

- ★ After disconnection, the current through R_1 REVERSES (it now carries the inductor's current, $4\,\mathrm{A}$ at $t = 0^+$).

Questions: JEE Advanced Level (3)

ideal L and C; long straight wire, $\mu_0 = 4\pi \times 10^{-7}\,\mathrm{T\,m\,A^{-1}}$

AS · JEE Advanced level · Matching

An ideal LC circuit: $L = 10\,\mathrm{mH}$, $C = 1\,\mu\mathrm{F}$, maximum charge $10\,\mu\mathrm{C}$. List I: (P) ω in $\mathrm{rad\,s^{-1}}$ (Q) maximum current in A (R) total energy in J (S) current in A when $q = 5\,\mu\mathrm{C}$. List II: (1) 0.0866 (2) 10^4 (3) 0.1 (4) 5×10^{-5}

Solution: $\omega = \frac{1}{\sqrt{10^{-2}\times 10^{-6}}} = 10^4$; $I_0 = \omega q_0 = 0.1\,\mathrm{A}$;

$U = \frac{q_0^2}{2C} = 5 \times 10^{-5}\,\mathrm{J}$; $|I| = \omega\sqrt{q_0^2 - q^2} = 0.1 \times \frac{\sqrt{3}}{2} = 0.0866\,\mathrm{A}$.

Answer: P–2, Q–3, R–4, S–1

A6 · JEE Advanced level · Numerical

A long straight wire carries $10\,\mathrm{A}$. A rod perpendicular to it, in a plane containing the wire, has its ends $0.1\,\mathrm{m}$ and $0.3\,\mathrm{m}$ from the wire and moves at $5\,\mathrm{m\,s^{-1}}$ parallel to the current. EMF (in $\mu\mathrm{V}$)? Which end is at higher potential?

Solution:

$\varepsilon = \frac{\mu_0 I v}{2\pi} \ln\frac{b}{a} = 2 \times 10^{-7} \times 10 \times 5 \times \ln 3 = 1.10 \times 10^{-5}\,\mathrm{V} = 11.0\,\mu\mathrm{V}$.

$\vec{v} \times \vec{B}$ points towards the wire: the NEAR end.

Answer: ≈ 11.0, the end nearer the wire

Golden points

- ★ LC: $|I| = \omega\sqrt{q_0^2 - q^2}$, like $v = \omega\sqrt{A^2 - x^2}$ in SHM.
- ★ Non-uniform B: integrate $B(x)\,v\,dx$; never use the field at the middle of the rod.